Yuna Watanabe

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RESEARCH INTEREST

My research centers on detecting and intervening in interpersonal relationship dynamics, with a focus on romantic couples. I leverage ambulatory physiological signals, including measures of physiological synchrony, to capture nuanced interpersonal dynamics in everyday life, combining statistical modeling and machine learning to extract meaningful patterns from complex multivariate data. My long-term goal is to empower couples, including neurodiverse couples, to reflect on and improve their relationships outside of clinical settings, supporting healthier and more resilient partnerships. Alongside this, I am committed to advancing the rigor of psychophysiological research by developing open-source algorithms and toolkits that improve the accuracy, validity, and reproducibility of ambulatory signal processing for the broader research community.

EDUCATION

Northeastern University Sept 2023–Present

- Ph.D. in Personal Health Informatics
- GPA: 4.00/4.00

University of Tokyo Apr 2019–Mar 2023

- B.S. in Arts and Sciences (Information Science major)
- GPA: 3.87/4.3

RESEARCH EXPERIENCE

Graduate Research Assistant | Northeastern University

Advisors: Prof. Matthew Goodwin and Prof. Varun Mishra Sept 2023–Present

- Develop and evaluate deep learning models (TCN, PatchTST, and ShapeNet) on multimodal ambulatory physiological data (accelerometer, PPG, EDA) to predict aggressive behaviors in autistic children.
- Designed and validated a machine learning pipeline for context-aware PPG preprocessing on wearable devices, reducing pulse-rate variability RMSSD estimation error by up to 279 ms compared with fixed baseline filtering and improving downstream stress detection accuracy.
- Design and execute a wearable benchmarking study (N=24) collecting multimodal physiological signals (PPG, ECG, EDA, and accelerometry) under stress protocols to evaluate sensor performance and model generalizability.
- Design and evaluate a probing technique for couples to surface their relationship needs by examining the congruence and discrepancy in their perceptions.

Undergraduate Research Assistant | University of British Columbia

Advisor: Prof. Karon MacLean May 2022–Apr 2023

- Investigated the impacts of haptic biofeedback on user's heart rate and the quality of online collaboration by conducting a study with 20+ participants

Undergraduate Research Assistant | The University of Tokyo

Advisor: Prof. Jun Rekimoto Apr 2022–Aug 2023

- Conducted a study with 20+ participants and examined the impact of haptic biofeedback on users' emotions and physiology

- Developed a Python-based open-source toolkit (OpenRBSync) to visualize interpersonal physiological synchrony in real time

Undergraduate Research Assistant | The University of Tokyo
Advisor: Prof. Daisuke Uriu Mar 2022–Mar 2023

- Designed and implemented an AR-based telepresence system and investigated its effects on communication

Undergraduate Research Assistant | The University of Tokyo
Advisor: Prof. Takuji Narumi Aug 2021–Mar 2022

- Examined the cognitive effects of a VR avatar that two users operate together
- Designed a study protocol, ran a user study with 30+ participants, and conducted statistical analysis as a project leader

PUBLICATIONS

[P1] Potter M, Everett M, Singh A, Stratis G, **Watanabe Y**, Demirkaya A, Erdogmus D, Imbiriba T, Goodwin MS. Temporal point process modeling of aggressive behavior onset in psychiatric inpatient youths with autism. arXiv preprint arXiv:2503.15821. 2025 Mar 20. Accepted at Nature Scientific Reports.

[P2] **Watanabe Y**, Yamane N, Sathyanarayana A, Mishra V, Goodwin MS. Beyond Motion Artifacts: Optimizing PPG Preprocessing for Accurate Pulse Rate Variability Estimation. In Companion of the 2025 ACM International Joint Conference on Pervasive and Ubiquitous Computing 2025 Oct 12 (pp. 1176-1182).

[P3] Dunn J, Mishra V, Shandhi MM, Jeong H, Yamane N, **Watanabe Y**, Chen B, Goodwin MS. Building an open-source community to enhance autonomic nervous system signal analysis: DBDP-autonomic. Frontiers in Digital Health. 2025 Jan 9;6:1467424.

[P4] Yazaki T, **Watanabe Y**, Kong L, Inami M. Design and field study of Syn-Leap: A symmetric telepresence system for immersion switching and walking across multiple locations. In Proceedings of the 22nd international conference on mobile and ubiquitous multimedia 2023 Dec 3 (pp. 353-365).

[P5] Yazaki T, Uriu D, **Watanabe Y**, Takagi R, Kashino Z, Inami M. "Oh, could you also grab that?": A case study on enabling elderly person to remotely explore a supermarket using a wearable telepresence system. InProceedings of the 22nd International Conference on Mobile and Ubiquitous Multimedia 2023 Dec 3 (pp. 340-352).

[P6] **Watanabe Y**, Cang XL, Guerra RR, Mclaren D, Vyas P, Rekimoto J, Maclean KE. Demonstrating Virtual Teamwork with Synchrobots: A Robot-Mediated Approach to Improving Connectedness. In Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems 2023 Apr 19 (pp. 1-4).

UNDER REVIEW SUBMISSIONS

[U1] **Watanabe Y**, Yamane N, Sathyanarayana A, Mishra V, Goodwin MS. A Context-Aware Framework for Optimal Filter Parameter Selection: Enhancing PRV Estimation in Wearable Wrist PPG Signals. Under revision at IMWUT'26.

[U2] Potter M, Everett M, Erdogmus D, **Watanabe Y**, Imbiriba T, Goodwin MS. Hierarchical Temporal Point Process Modeling of Aggressive Behavior Onset in Psychiatric Inpatient Youth with Autism for Branching Factor Estimation. arXiv preprint arXiv:2507.12424. 2025 Jul 16. Under review at BMC Medical Research Methodology.

AWARDS

Khoury PhD Network Travel Award Oct 2025

- Awarded $500

Japan Student Services Organization Scholarship Sept 2025–Aug 2028

- Awarded $30,000/year
- Japanese government fellowship

Nokia Bell Labs OmniBuds Open Research Call Jun 2025

- Received testing devices for approved research proposal

Grad Cohort for Women Apr 2024

- Selected as a student for the Grad Cohort for Women Workshop 2024

Heiwa Nakajima Scholarship Sept 2023–Aug 2025

- Awarded $24,000/year
- Acceptance rate 5%

TEACHING EXPERIENCE

Teaching Assistant Jan 2024–Apr 2024

- Principles of epidemiology in medicine & public health

STUDENT MENTORING

Hanrong Zhang	| MS in Computer Science	2026–Present
Amrutha Arunkumar	| MS in Personal Health Informatics	2026–Present
Olivia Ragheb	| MS in Physician Assistant	2025–Present
Akshara Reddy Patlannagari	| MS in Data Analytics Engineering	2025–Present
Kim Chuong Tran Dang	| BS in Data Science and Behavioral Science	2025–Present
Haifa Alhokair	| BS in Psychology	2025–Present

SERVICE

Reviewing	CHI LBW (2025)

SKILLS

Programming	Python, C++, C#, JavaScript, TypeScript, SQL, MATLAB, R, Swift, HTML, CSS
Frameworks and Libraries	React, D3, PyTorch, Firebase
Tools	Git, Unix/Linux, Docker, Shell scripting